Luiss
Libera Università Internazionale
degli Studi Sociali Guido Carli

Exercises: math and datetime modules

Introduction to Computer Programming

Management and Artificial Intelligence

`math` module

`sqrt()`

```python
import math

number = 25
sqrt_value = math.sqrt(number)
print(f"The square root of {number} is {sqrt_value}")
```

```
The square root of 25 is 5.0
```

`pi` and `sin()`

```
angle_in_radians = math.pi / 6  # 30 degrees
sine_value = math.sin(angle_in_radians)
print(f"Sine of 30 degrees (in radians) is {sine_value}")
```

```
Sine of 30 degrees (in radians) is 0.49999999999999994
```

factorial()

```
num = 5
fact_value = math.factorial(num)
print(f"The factorial of {num} is {fact_value}")
```

```
The factorial of 5 is 120
```

log()

```
number = 10
natural_log = math.log(number)
print(f"Natural logarithm of {number} is {natural_log}")
```

```
Natural logarithm of 10 is 2.302585092994046
```

pow()

```
base = 2
exponent = 3
power_value = math.pow(base, exponent)
print(f"{base} raised to the power of {exponent} is {power_value}")
```

```
2 raised to the power of 3 is 8.0
```

`gcd()`

```python
num1 = 48
num2 = 180
gcd_value = math.gcd(num1, num2)
print(f"The GCD of {num1} and {num2} is {gcd_value}")
```

```
The GCD of 48 and 180 is 12
```

`degrees()`

```python
radians_value = math.pi
degrees_value = math.degrees(radians_value)
print(f"{radians_value} radians is equal to {degrees_value} degrees")
```

```
3.141592653589793 radians is equal to 180.0 degrees
```

`hypot()`

```
side_a = 3
side_b = 4
hypotenuse = math.hypot(side_a, side_b)
print(
    f"The hypotenuse of a right triangle with sides {side_a} and {side_b} is {hypoter
)
```

```
The hypotenuse of a right triangle with sides 3 and 4 is 5.0
```

`ceil()`

```
num = 4.3
ceil_value = math.ceil(num)
print(f"The ceiling of {num} is {ceil_value}")
```

```
The ceiling of 4.3 is 5
```

`floor()`

```
num = 4.8
floor_value = math.floor(num)
print(f"The floor of {num} is {floor_value}")
```

```
The floor of 4.8 is 4
```

Exercises

- Notebook with exercises on math (no hints)
- Notebook with exercises on math (with pseudocode)
- Notebook with exercises on math (with solutions)

`datetime` module

`now()`

```python
from datetime import datetime

now = datetime.now()
print("Current date and time:", now)
```

```
Current date and time: 2024-09-17 21:03:51.704489
```

`today()`

```
today = datetime.today()
print("Year:", today.year)
print("Month:", today.month)
print("Day:", today.day)
```

```
Year: 2024
Month: 9
Day: 17
```

`datetime()`

```
specific_date = datetime(2024, 12, 25, 15, 30)
print("Specific date and time:", specific_date)
```

```
Specific date and time: 2024-12-25 15:30:00
```

`strftime()`

```
formatted_date = today.strftime("%Y-%m-%d %H:%M:%S")
print("Formatted date:", formatted_date)
```

```
Formatted date: 2024-09-17 21:03:51
```

`strptime()`

```
date_string = "2024-09-17 14:30:00"
parsed_date = datetime.strptime(date_string, "%Y-%m-%d %H:%M:%S")
print("Parsed date:", parsed_date)
```

```
Parsed date: 2024-09-17 14:30:00
```

`timedelta()`

```python
from datetime import timedelta

future_date = today + timedelta(days=10)
past_date = today - timedelta(days=5)
print("Date 10 days from now:", future_date)
print("Date 5 days ago:", past_date)
```

```
Date 10 days from now: 2024-09-27 21:03:51.715959
Date 5 days ago: 2024-09-12 21:03:51.715959
```

`datetime` subtraction

```
date1 = datetime(2023, 9, 1)
date2 = datetime(2024, 9, 1)
difference = date2 - date1
print("Difference between two dates:", difference.days, "days")
```

```
Difference between two dates: 366 days
```

`time()`

```
current_time = now.time()
print("Current time:", current_time)
```

```
Current time: 21:03:51.704489
```

Check if an year is a leap year

```
year = 2024
is_leap = year % 4 == 0 and (year % 100 != 0 or year % 400 == 0)
print(f"Is {year} a leap year?", is_leap)
```

```
Is 2024 a leap year? True
```

weekday()

```python
weekday = today.weekday()  # Monday = 0, Sunday = 6
print(f"Today is day {weekday} of the week (0=Monday, 6=Sunday)")
```

```
Today is day 1 of the week (0=Monday, 6=Sunday)
```

Exercises

- Notebook with exercises on math (no hints)
- Notebook with exercises on math (with pseudocode)
- Notebook with exercises on math (with solutions)